

Biology

Chapter 2: Human Anatomy

Welcome to AnithUncommon's Sample Resources!

We are assuming you are reading this as a student. So What will **YOU** Expect From This Sample?

In Biology, Human Anatomy, Ch. 2.1

This sample will be divided into **three** parts: Introduction to Samples (this part), Lesson Objectives (What is Expected From Students), and most crucially, Syllabus Contents.

Note that the 'Syllabus Content' does NOT stand as a complete syllabus itself; rather, it is a tight compilation of notes gathered by Azalia to create a comprehensive sample for students.

If you want a complete version of these notes and think that you need help in Biology, join our nonprofit!

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This sample is made thanks to Azalia.

— Made by students, for students.

Lesson Objectives:

At the end of the lesson, students should be *able to...*

1. **Outline** the **structure of sensory, relay, and motor neurones** and **explain** how their features help the rapid conduction of impulses.
2. **Describe** the **role of sensory receptor cells** in detecting stimuli and converting various forms of energy into electrical impulses.
3. **Explain** how a **nerve impulse is transmitted** along an axon, including the stages of an action potential, including depolarization, repolarization, and hyperpolarization.

Reflective Question to Consider:

The human body is an intricately organized structure where cells, tissues, and organs work together. There are approximately 11 organ systems building our body. Anatomy focuses on understanding these structural levels and how they support the body's overall function.

Below are the following core questions you will face as we explore this chapter:

- How do the different structures and positions of the organ system help in maintaining the whole condition?
- In which way those structures communicate even though they're spread from head to toe?

Unit 2: Human Anatomy

Pro Tip!

- Memorizing the whole systems can be overwhelming! But what we will learn in this chapter is all about us, all about our bodies that we're in. Put your mind into it, feel every heartbeat and every breath. This will really help you to think also to feel what mechanism is happening in the body.

Activities in Class:

- Lesson Breakdown
- Kahoot-Based Review for deep memorizing
- Analyzing real-world conditions related to the body

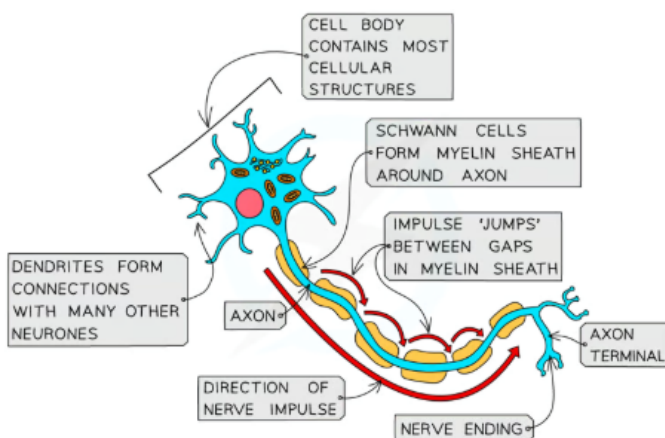
2.1 Communication in A System

Neurones

A. What Are Neurones?

They are **specialized cells** which **carry electrical impulses** around the body. Just think of them like pipes that carry the flow of water to the designated place.

B. Structure



Mentor's Notes ★

The three types of neurones have their own position that reflects their functions.

Remember this little pathway of them:

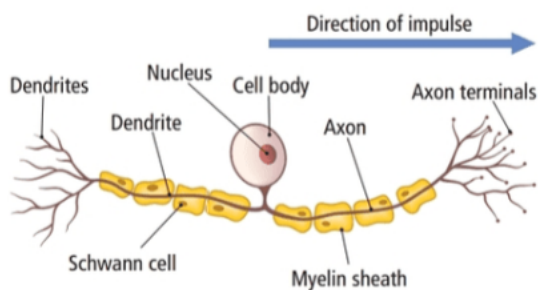
Sensory (hearing, touching, or anything that the five senses could feel) → Relay (processing the info that sensory detected, signalling to other) → Motor

Most of them have these three main parts: **dendrite, cell body and axon**. They're not always visible in each type, but these three are the prominents:

- **Dendrite:** The upper part of neurone, shaped like a root. Dendrite is useful in **receiving signals** from other neurones and **conveying the neurochemical reaction** to the cell body.
- **Cell body/Soma:** The middle part that **contains the nucleus** (the circle one), useful in **processing the information** and **manages the flow of reaction**.
- **Axon:** The long-entailed part of the neurone, stretches from the **Axon Hillock (the cell body-axon junction)** to the claw-shaped **end tail**. This part **conducts electrical impulses away** from the cell.

C. Types and Features

a. Sensory Neurone



→ A type of neurone that carries impulse from **receptors to the Central Nervous System (CNS;** could be **brain or spinal cord**).

→ The structure is distinct: **dendrite** and **the cell body don't combine**. The cell body **stands alone in the middle** of the neurone, while the dendrite is still

(motor means movement, what parts of body can we move? Hands, feet, etc)

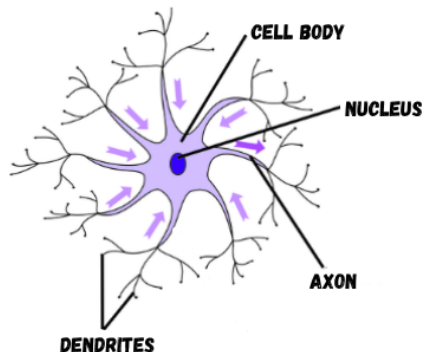
Why is the relay neurone not insulated by myelin sheath? While the rest are insulated to speed up the transmission, the relay neurone doesn't require any since the axon is short. Both sensory and motor neurone have a long axon, so they will need something to speed up the message sending.

positioned in **the upper part** of the neurone.

→ The sponge-like bandage along the tail is called **myelin sheath**, made out of **Schwann cells**. These bandages **maximize the speed** of transmission.

→ Often found in sensory organs; **the five senses**.

b. Relay Neurone



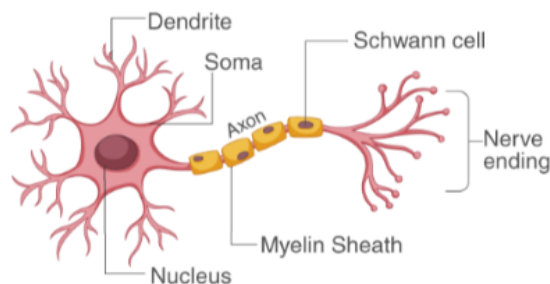
→ A type of neurone that sits **within the CNS** to **connect** between **sensory and motor neurones**.

→ All three prominent parts are **not necessarily noticeable**, but yes, they exist. All is **combined** into a multi-armed star-shaped neurone.

→ Relay neurone is the only neurone that has **no myelin sheath** along the axon.

→ Found within the **brain** and **spinal cord**.

C. Motor Neurone



→ A type of neurone that carries impulse from **CNS to effectors** (like hands or feet when they receive command from the brain).

- Have a normal and visible structure: dendrite, cell body and axon.
- Like the sensory neurone, they're also **insulated by myelin sheath** for quick transmission.
- They stretch from the **CNS to any muscle** in the body as **the effectors**.

Sensory Receptor Cells

A. What are Sensory Receptor Cells?

Sensory receptor cells are a type of cells that **catch a stimulus** and **convert** them into an **electrical impulse**. The feature of **converting stimuli** to impulse is also called transducing, so basically they're **transducers**. They can only react to **specific stimuli** that **match** their **functions**.

B. Receptors and The Stimuli Table

Sensory Receptor	Stimulus
Photoreceptor	Light
Chemoreceptor	Chemicals
Mechanoreceptor	Mechanical stretching
Baroreceptor	Blood pressure
Osmoreceptor	Body fluids

Transmission Of Impulse

A. What This Actually Means

The term "impulse transmission", in science, specifically refers to the **change in the amount** of **Na⁺ and K⁺** ions along the **axon**, neither a flow of electrons nor fluid.

Sensory receptor cells are those we find in the nerve ending of sensory neurones. They mostly spread in the five senses or internal organs as an internal alarm.

Types of receptors are named based on the Latin terms for their key stimuli:

Photo → Light
(When we're taking photos, all we need is light, right? That's why it's called *photography*)

Chemo → Chem

Mech → Mechanic

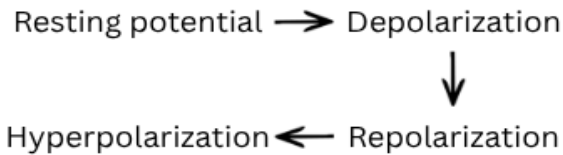
Baro → Pressure

(Heard of *barometer*? It's to measure the pressure/weight of the air)

Osmo → Fluids (you must have heard of *osmoregulation*, which means fluid regulation)

B. Order of Events in Action Potential

Talking about impulse transmission, the conduction of electrical messages itself is called **action potential**. This action potential has a specific sequence of events happening in each transmission:



C. How It Is Done

a. Resting Potential

→ The **normal condition** of the nerve at any times beyond action potential.

→ Marked by more **Na⁺ outside** and **K⁺ inside** the axon, making the **inside** more **negatively-charged** (**=-70mV**)

→ To keep this condition, the **sodium-potassium** active pump **lets in 2 K⁺** and **exports 3 Na⁺ out**.

b. Depolarization

→ Means the **opening of Na⁺ channels** that are spread **along the axon in turns** as a **response** to a stimulus received by dendrites.

→ The **Na⁺** that are supposed to be outside the axon, **flows inside** in a huge amount, making the inside **more positive** (minimum until **-55mV** so they can continue the transmission to the end).

→ If they've reached **-55mV**, the **Na⁺** will flow **much higher** inside, making the charge **inside** now **+30mV**.

c. Repolarization

→ After about **1 millisecond** (ms) after depolarization, **all Na⁺ channels are closed** to prevent more flow of **Na⁺**.

→ Vice versa, the **K⁺ channels** along the axon **open** to

When understanding the action potentials, keep in mind that in this context, we seek *balance*. When both positive ions meet in depolarization, the charge is completely positive. But, as the nature of the axons are "different" in charge, one of them needs to leave. The axon can't stand too many guests. So, the K⁺ goes, leaving the Na⁺ alone holding the negativity inside the axon. There we got the initial balance.

Polar here refers to the way the inside and outside of the axon is different in charge. Simply, like how one is (+) and one is (-).

The term

let the K^+ leaves the axon, making it back to the initial negativity (approx. $-70mV$)

d. Hyperpolarization

→ Sometimes, the K^+ overdid it. It leaves the cell too much, resulting in the insides being a bit more negative than normal (around $-80mV$ to $-90mV$). But, this usually lasts only for a short time.

All of these phases of action potentials last only about **1 to 2 milliseconds**, which makes it even faster than when we're blinking.

And last note, the flow of action potential **will always be from dendrites, along the axon to the nerve ending**. They will **never be vice versa**.

"depolarization", "repolarization", and "hyperpolarization" are named to the essence of the phase:

De: making it less, means making the difference between two sides less, by letting Na^+ rushes into the axon.

Re: reviving something, means returning the charge difference both inside and outside the axon.

Hyper: too much, means after reviving the negativeness of axon insides, they become a bit too negative from the usual threshold ($-70mV$)

Mentors' Sources:

Video link to make a clearer visualization of impulse transmission:

[The Nerve Impulse](#)

[How Nerve Communicate](#)

Save My Exams Notes:

[Action Potentials](#)

[Transmission of Nerve Impulse](#)